

Rubik's Cube Solvability using Group Theory

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April 2026

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1 Introduction

In 1878, puzzle maker Sam Loyd released a 15-square puzzle with a prize of \$1000 for the first person to solve it. Legend goes that "people became infatuated with puzzle" [Lau03] in hopes of winning the money. In this pursuit, "pilots are said to have wrecked their ships, and engineers rush their trains past their stations" [Lau03], people would claim to have solved the puzzle only to mysteriously fail to recollect the exact sequence of moves that solved the puzzle. Ultimately, no one was ever able to claim the prize from Loyd for one simple reason: the puzzle was unsolvable!

Using group theory ideas, we will explore the Rubik's Cube Group for both the 3x3 and 2x2 Rubik's Cubes. We'll also discuss an algorithm to solve the cube, why it works, and how to make impossible Rubik's Cubes.

2 Group Theory in the Rubik's Cube

2.1 Notation

We will represent each cube using Singmaster notation, which represents a clockwise (cw), 90 degree turn of each side of the Rubik's cube using a letter, with respect to a stationary observer [Mula]:

R	90 deg. cw rotation of the RIGHT face
L	90 deg. cw rotation of the LEFT face
U	90 deg. cw rotation of the UP face
D	90 deg. cw rotation of the DOWN face
F	90 deg. cw rotation of the FRONT face
B	90 deg. cw rotation of the BACK face

Table 1: Table of the 6 Basic Cube Moves

Note that R', L', U', D', F', B' represent the inverse of each cube move, or a 90 degree counter clockwise (ccw) rotation of the face.

2.2 The Rubik's Cube Group.

Let's briefly discuss how the set of all possible moves of the Rubik's Cube form a group: $(G, \cdot)$. Let M_1 and M_2 be two moves within the set of possible Rubik's Cube moves, and let $\cdot$ be defined as performing M_1 followed by M_2 . In order to be a group, the Rubik's Cube must meet the following 4 criteria:

- **Closure.** Under $\cdot$, since M_1 and M_2 are possible Rubik's Cube moves, then so is $M_1 \cdot M_2$. Therefore, $M_1 \cdot M_2 \in G$ and G is closed.

- **Identity elements.** Let e be the "nothing" move, were we do nothing to the Rubik's Cube. Then $e \in G$ and for any $M \in G$, $e \cdot M = M \cdot e = M$, and e is the identity element.
- **Associativity.** Clearly, G is associative under, $\cdot$. Let $M_1, M_2, M_3 \in G$, we can see that $(M_1 \cdot M_2) \cdot M_3 = M_1 \cdot (M_2 \cdot M_3)$, because the 6 basic moves that each M_i consists of are associative. Therefore, G is associative under $\cdot$.
- **Inverses.** Finally, in order for G to be a group, there must exist an inverse for every Rubik's Cube move. Earlier, we defined for each basic move X' as it's inverse. So, we can define for $M \in G$, define M as $M := (X_1 \cdot X_2 \dots X_n)$, for an arbitrary move, where $X_1, X_2, \dots X_n$ are each one of the 6 basic moves. Then, we can define $M' = (X_1 \cdot X_2 \dots X_n)^{-1} = X'_n \cdot X'_{n-1} \dots X'_1$. This is the inverse of M , because

$$X'_n \cdot X'_{n-1} \dots X'_1 \cdot (X_1 \cdot X_2 \dots X_n) = X'_n \cdot X'_{n-1} \dots X'_1 \cdot (X_1 \cdot X_2 \dots X_n) = e$$

(for a more formal proof by induction see [Che]).

Now, that we have established that $(G, \cdot)$ is a group, we can move on to more interesting explorations. It is also important to note that the group of Rubik's Cube moves is the same for both the 3x3 and 2x2 Rubik's Cube. For $n \times n$ Rubik's Cubes with $n > 3$, we would need to define more moves for the interior, movable cubes, so we will only be discussing the 3x3 or 2x2 Cubes here.

2.3 Showing G is Non-Abelian.

We can do this with a counter example. Consider $R \cdot U$ and $U \cdot R$. Performing each of these moves on the Pocket Cube in the solved state, results in the states in Figure 1 and Figure 2, respectively. Clearly, $RU \neq UR$. Therefore, G is non-abelian in the 2x2 case.

This result generalizes to the 3x3 Rubik's Cube as well, since the Pocket Cube represents the corners of the 3x3 Rubik's Cube. Therefore, G is non-abelian in the 3x3 Rubik's Cube case as well.

2.4 Generators.

Because we have shown that G is not abelian, we know that it is **not cyclic**. That is, there is no $g \in G$ such that $\langle g \rangle = \{g^n : n \in \mathbb{N}\}$ [Lau03]. But, there are some generating sets. An obvious generator is the set of our 6 basic moves (plus the "nothing move") with their inverses, $\langle e, L, R, U, D, F, B, L', R', U', D', F', B' \rangle$. Maybe less is obvious is that just the 6 basic moves themselves constitute a generator of G [Gar11], that is:

$$\langle L, R, U, D, F, B \rangle = G$$



Figure 1: RU permutation.

we can "undo" these using combinations of the other five moves (i.e., we can generate X' for any $X \in \{L, R, U, D, F, B\}$ using combinations of the other moves) [Gar11]. Also, observe that $e = LLLL$, for example.

Using the previous observation, it becomes more obvious that any one of the 6 basic moves can be generated from the other 5. That is,

$$G = \langle L, R, U, D, F \rangle$$

$$G = \langle L, R, U, D, B \rangle$$

$$G = \langle L, R, U, F, B \rangle$$

$$G = \langle L, R, D, F, B \rangle$$

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$$G = \langle R, U, D, F, B \rangle$$

are all generating sets, and $(G, \cdot)$ is **five-generated** [Gar11]. An example from Garron's paper is that

$$D = (L' R F F B B L' R) U (L' R F F B B L' R)$$

Further more, G is **two-generated**. An example from Garron is

$$G = \langle U B L U L' U' B', R R F L D' R' \rangle$$



Figure 2: UR permutation.

2.5 Subgroups.

Recall that a subgroup, H , is a subset of a group that is a group itself under the same operation [Lau03]. This written as $H \leq G$. In fact, $(G, \cdot)$ itself is a subset of both S_{54} and S_{48} [Bob].

Recall that the symmetric group, S_n consists of all permutations of a set with n elements. In the case of S_{54} , we can see that the 3x3 Rubik's Cube has 54 facelets that can be permuted, but as we already discussed and will delve into deeper later, not all these permutations are possible, but G is a group on it's own. Therefore, $G \leq S_{54}$. The case for S_{48} is similar, except here we observe that that center facelet on each face of the Rubik's Cube never move, leaving only 48 facelets to permute. Again, not all permutations of these 48 facelets are feasible, so it follows that $G \leq S_{48}$ [Bob].

Similarly, the Pocket Cube is $G \leq S_{24}$. It has 24 facelets that can be permuted, but not all these permutations are feasible.

More interesting are the subgroups of $(G, \cdot)$, Garron [Gar11] lays out many examples, here we will discuss some of the most interesting. The obvious subgroups are [Lau03]:

- $G \leq G$
- $e \leq G$

Unfortunately, because $(G, \cdot)$ is not cyclic, the number and orders of the subgroups isn't obvious. So, we'll just discuss a few interesting ones,

First, for any $g \in G$, $\langle g \rangle \leq G$ [Gar11]. Garron doesn't prove this in his paper, so we will show it here:

- **Subset.** $\langle g \rangle \subset G$, since $g \in G$ and G is closed under $\cdot$.
- **Closure.** Because $\langle g \rangle = \{g^n : n \in \mathbb{N}\}$, any $g_1, g_2 \in \langle g \rangle$ take the form $g_1 = g^n$, $g_2 = g^m$. Therefore, $g_1 \cdot g_2 = g^{n+m} \in \langle g \rangle$ by definition of a generator, so $\langle g \rangle$ is closed under $\cdot$.
- **Identity element.** In order for $\langle g \rangle$ to be a subgroup of $(G, \cdot)$, $e_G \in \langle g \rangle$. Recall that e_G is the "do nothing move". We know that repeating the same move over and over will eventually result in original state of the cube [Gar11]. This means $\exists k$ such that $g^k = e_G \forall g \in G$. Since $g^k \in \langle g \rangle$, it follows that $e_G \in \langle g \rangle \forall g \in G$.
- **Inverses.** Recall that for an element $x \in \langle g \rangle$ the inverse is $x' \in \langle g \rangle$, such that $xx' = x'x = e_G$. For all $x \in \langle g \rangle$, let $x' = g^{k-1}$, where k is such that $g^k = e_G$ from the proof of the Identity Element. Then, $xx' = x'x = e_G$ becomes

$$g \cdot g^{k-1} = g^{k-1} \cdot g = e_G$$

which holds $\forall g \in G$. Therefore, the inverses of all elements exist in $\langle g \rangle$.

- **Associativity.** Associativity is inherited from G .

Therefore we have proven that for any legal move $g \in G$, $\langle g \rangle \leq G$. Unlike G , these subgroups are clearly abelian, by the commutativity of scalar multiplication. Note that this mean that the generators we discussed in Section 2.4 Generators are all also subgroups of G .

2.6 Normal Subgroups of G .

According to the Rubik's Cube Group Wikipedia page [Wik], the group C_o , or the subset of moves that leaves every cubelet fixed, but changes their orientations is a normal subgroup. They omit the proof of this, so we will verify it here. First, let's make sure $C_o \leq G$ by making sure it fulfills the following criteria:

1. **Subset.** Clearly, $C_o \subset G$, by definition since these are legal moves that also have the special property of leaving the cubelets fixed, but rotating the facelets.
2. **Identity element.** Recall e_G is the empty move. Therefore, $e_G \in C_o$ since the cubelets are left fixed by e_G .
3. **Closure.** Let $A, B \in C_o$, then $AB \in C_o$ because neither A nor B move any cubelets, and therefore their composition AB also will not move any cubelets.

4. **Inverses.** As aforementioned, all $g \in G$ can be written as the composition of the 6 basic moves: R, L, U, D, F, B or their inverses. This is also true for C_o . Then, $g \in C_o$ can be expressed as $g = A_1 A_2 \dots A_n$ where $A_i \in R, L, U, D, F, B$ or their inverses. Therefore, $\exists g' \in C_o$, where $g' = A'_n A'_{n-1} \dots A'_1$, which will also not permute any cubelets.

Because C_o meets these criteria, we can confirm $C_o < G$. Now, we know that a normal subgroup [Lau03] is such that $\forall c \in C_o, g \in G$

$$gC_o g' = \{gcg' | c \in C_o\} = C_o$$

How do we show this is true? Let's define the state of our 3x3 Rubik's Cube cubelets as

$$g = \begin{bmatrix} 1 & 2 & \dots & 26 \\ a_1 & a_2 & \dots & a_{26} \end{bmatrix}$$

where a_i is the current cubelet occupying cubicle i . As we will see later we can write every move $g \in G$ as a bijective map, call it θ , in the form of a permutation, like

$$g = \begin{bmatrix} 1 & 2 & \dots & 26 \\ \theta(1) & \theta(2) & \dots & \theta(26) \end{bmatrix}$$

This just moves the cubelets around the Rubik's Cube (we will ignore the facelets for now, since C_o is only concerned with fixed cubelets). Clearly,

$$g' = \begin{bmatrix} \theta(1) & \theta(2) & \dots & \theta(26) \\ 1 & 2 & \dots & 26 \end{bmatrix}$$

and

$$gg' = e$$

which means all the cubelets are left fixed. Since $\forall c \in C_o$ the map will look like the cubelet-level identity move, e ,

$$c = e = \begin{bmatrix} 1 & 2 & \dots & 26 \\ 1 & 2 & \dots & 26 \end{bmatrix}$$

since by definition none of the cubelets will move for moves $c \in C_o$, we are left with $\forall c \in C_o, g \in G$

$$gcg' = geg' = gg' = e$$

and as we already established

$$e \in C_o$$

Therefore, $\forall c \in C_o, g \in G$

$$gcg' = e \in C_o$$

and we can conclude

$$C_o \triangleleft G$$

3 The Fundamental Theorem of Cube Theory.

3.1 Cube Orientation.

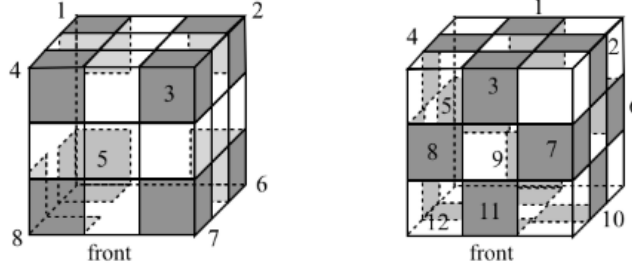


Figure 3: Visualizing the 8 corner positions and 12 edge positions on the 3x3 Rubik's Cube [Mula].

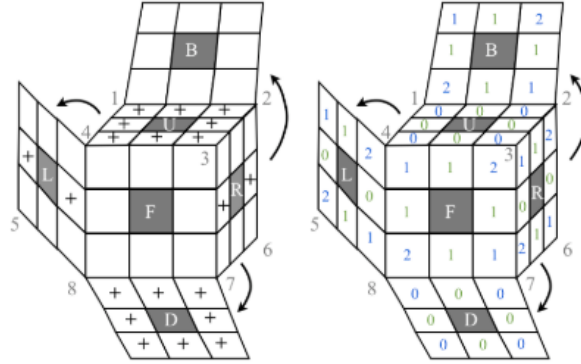


Figure 4: Visualizing the orientation for the corner and edge facelets [Mula].

Borrowing the orientation graphics for Mulholland's webpage, we can see how we are labeling each cube. The corner cubes can be positioned in 8 different positions, often called, "cubicles", labeled 1 through 8. Relative to the user, the cubicles are not mutable, but the cube occupying that cubicle is. For example, cubicle 3 will always be the top-right cubicle to the user, but any of the 8 corner cubes may be filling that position (see Figure 3). This extended to labeling the 12 edge cubicles.

We also need to keep track of how each corner and edge piece are oriented within their current cubicles. To facilitate this, we will need to do two things:

1. Label each facelet of each corner cube as follows: (a) in the solved position,

the facelets Upper and Downward faces are labeled as the "0" face, and (2) following each cube clockwise, label the other facelets "1" and "2", respectively.

So, when we discuss the orientation of the cube, each of the 3 possible orientations of cube are described by which facelet is facing the Upper or Downward facelet. For example, after move F , the yellow-green-red corner cube is now in cubicle 3, with orientation 2 (see 5). For reference, Figure 6

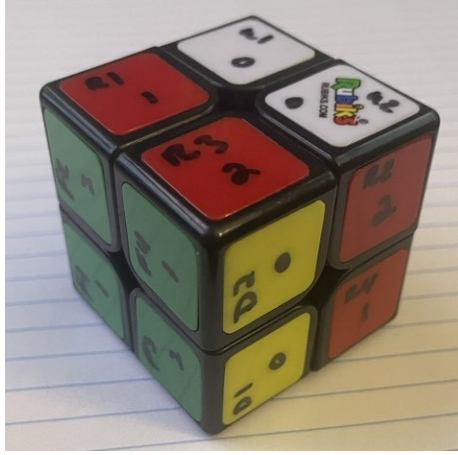


Figure 5: Pocket Cube after executing B .

3.2 Alternating Groups & Cycles.

3.3 FTCT: 3x3 Rubik's Cube.

We can represent the state of our 3x3 Rubik's Cube using a 4-tuple, (ρ, θ, v, w) , where:

- $\rho \in S_8$, $\rho(i) = j$ represents corner i in position j
- $\theta \in S_{12}$, $\theta(i) = j$ represents edge i in position j
- $v \in \mathbb{Z}_3^8$, represents the orientation of corners
- $w \in \mathbb{Z}_2^{12}$, represents the orientation of the edge pieces.

The 4-tuple (ρ, θ, v, w) corresponds to a **possible** configuration of our Rubik's Cube if and only if [Bob][Mulb]:

- Equal parity of permutations:

$$\text{sgn}(\rho) = \text{sgn}(\theta)$$



Figure 6: Solved Pocket Cube.

- $v_1 + v_2 + \dots + v_8 = 0 \pmod{3}$
- $w_1 + w_2 + \dots + w_{12} = 0 \pmod{2}$

In the FTCT, (1) says that there must be equal parity between the permutations of the corners and edges ("twists" in orientation), (2) is the conservation of corner twists: for every corner that flips clockwise, another flips counter clockwise, and (3) is conservation of the edge twists [Swe22]. Intuitively, this makes quite a bit of sense!

To expand on the group theory, here we can represent the group of **legal Rubik's Cube positions** as

$$(\rho, \theta, v, w) \in S_8 \times S_{12} \times \mathbb{Z}_3^8 \times \mathbb{Z}_2^{12}$$

which is the direct product between all these groups. Keep in mind the symmetric groups that the corners and edge pieces constitute they will come in handy in the section on the solving algorithm and how it works.

3.4 FTCT: Pocket Cube.

For the Pocket Cube, because we have no edge pieces to worry about, we can represent or state with the 2-tuple, $(\rho, v) \in S_8 \times \mathbb{Z}_3^8$ and corresponds to a **possible** configuration of our Pocket Cube if and only if $v_1 + v_2 + \dots + v_8 = 0 \pmod{3}$. There are no edge pieces and parity of permutations to worry about for them.

3.5 Cardinality.

Normally, calculating the order of a group is trivial for small, finite groups because we can simply count the elements in our group. But, as we'll see, there's not enough time in a lifetime to count to $|G|$ for a 3x3 Rubik's Cube, and we needed to get creative!

Recall that in the FTCT we defined the groups that our corner positions, corner orientations, edge positions, and edge orientations were in

$$\rho \in S_8$$

$$\theta \in S_{12}$$

$$v \in \mathbb{Z}_3^8$$

$$w \in \mathbb{Z}_2^{12}$$

respectively. So, we can represent the **Illegal Cube Group** as the direct product of the groups

$$S_8 \times S_{12} \times \mathbb{Z}_3^8 \times \mathbb{Z}_2^{12}$$

From Lauritzen, we know that the order of a product group is given by $G \times H = |G||H|$. Therefore,

$$|S_8 \times S_{12} \times \mathbb{Z}_3^8 \times \mathbb{Z}_2^{12}| = (8!)(12!)(3^8)(2^{12})$$

which is the same as the number of all permutations of the Rubik's Cube we found earlier, regardless of legality.

- **Group order for the 3x3 Rubik's Cube.** Using the FTCT, we can compute $|G|$. Relating back to the FTCT, $|\mathbb{Z}_3^8| = 3^8$, but the sum of the corner orientations, $v_1 + \dots + v_8$, needs to be $\equiv 0 \pmod{3}$, which only happens for $\frac{1}{3}$ of all possible sums. Similarly, $|\mathbb{Z}_2^{12}| = 2^{12}$, but the sum of the edge orientations, $w_1 + w_2 + \dots + w_{12} \equiv 0 \pmod{2}$, which only happens for $\frac{1}{2}$ of the sums. Finally, $\text{sgn}(\rho) = \text{sgn}(\theta)$, clearly, this only happens $\frac{1}{2}$ of the time as well. So, only $\frac{1}{3}(\frac{1}{2})(\frac{1}{2}) = \frac{1}{12}$ of the total permutations are feasible [Bob]. Using our result from earlier:

$$|G| = \frac{1}{12} |S_8 \times S_{12} \times \mathbb{Z}_3^8 \times \mathbb{Z}_2^{12}| = \frac{1}{12} (3^8(8!)(2^{12})(12!)) = 43,252,003,274,489,856,000 \approx 43 \times 10^{18}$$

Which is a pretty big group! For a person counting elements in this group at a rate of 3 elements per second, it would take them over 400 billion years to count them all! (457,170,675,572 years, to be exact).

Note that by Lauritzen, we know that if $|G|$ is prime, then G is cyclic. But, we showed previously that G is not cyclic, which therefore means $|G|$ is not prime, which is obvious since it ends in a 0, but still worth noting!

- **Group order for the Pocket Cube.** Again, we know we can represent the cube moves of the Pocket Cube as the direct product of the following groups:

$$|S_8 \times \mathbb{Z}_3^8| = (8!)(3^8)$$

Using the same logic for the corners, but ignoring the the edges since the Pocket Cube doesn't have any. Also, because the Pocket Cube is essentially only the corners of the 3x3 Rubik's Cube, only 7 of the corners can be rotated independently and we divide by 24 to account for the lack of fixed centers ($3(8) = 24$, where there are 3 orientations and 8 locations for the static cube to be.) [Sch]

$$|G| = \frac{1}{24} \cdot \frac{1}{3}(8!3^8) = 3,674,160$$

possible configurations for the Pocket Cube—quite a few less than the 3x3 Rubik's Cube, but still way too many to count!

- **Element & Subgroup orders.** Because we have shown previously that for any $g \in G$, $\langle g \rangle \leq G$, we know that $|g| = |\langle g \rangle|$. Furthermore, from Garron, we know that there are 73 possible orders of subgroups, ranging from 1 to 1260. The trivial subgroup has order 1, $|\{e\}| = 1$, and an example of an element with order 1260 is $|RFFB'U'B'| = 1260$ [Gar11].

4 The 2 x 2 Rubik's Cube: The Pocket Cube

For sake of simplicity, we are going to focus our explorations on the 2x2 Rubik's Cube, otherwise known as the Pocket Cube. We can think of the Pocket Cube as just the 3x3 Rubik's Cube's corners. This means we can apply what we know about permuting the 3x3 Rubik's Cube's corners, without having to worry about permutation any edge cubies or center cubies. Before we get into solving, we'll just note again that the Pocket Cube is simply the corners of the regular 3x3 Rubik's Cube. So, we will label our cubicles the same way (with respect to the user) as in Figure 3. When we talk about the some of the group theory (particularly cycles) behind this algorithm, it will also be helpful to refer to Figure 7 for the name of each facelet, as well as Figure 8, which shows a 3-D Pocket Cube with it's facelets labeled.

5 How to Solve the Pocket Cube.

5.1 The "Quickest" Method.

With commutators and conjugates, we now have enough tools to solve the Rubik's Cube. The Pocket Cube only requires 2 commutators to solve. We'll discuss that algorithm here, and talk about the group theory and abstract algebra behind how it works.

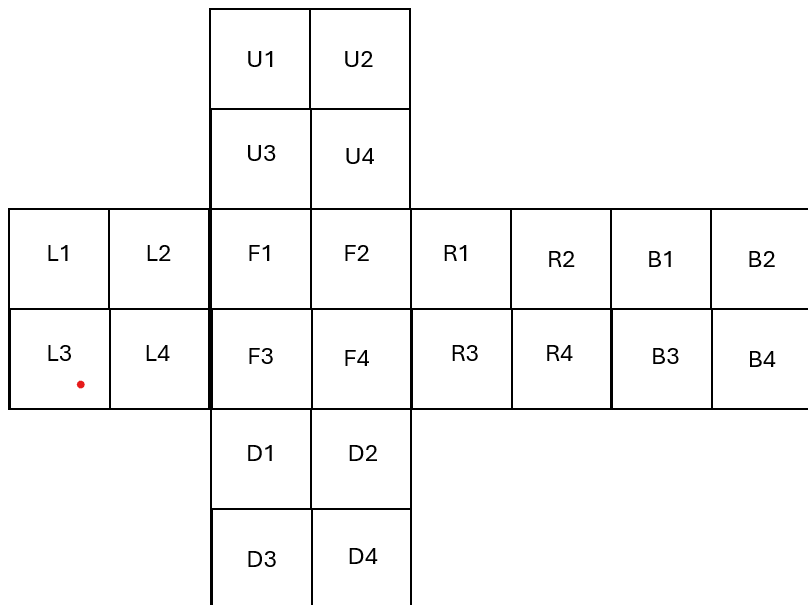


Figure 7: Labeled facelets for Pocket Cube.

5.2 Algorithm.

For the sake of clarity in this algorithm, we are going to assume that a white facelet is occupying D1. There is mention of something called **commutators** and **conjugates** in this algorithm. These are just moves that take a special form and have useful properties when solving a Rubik's Cube. We will discuss how and why these work in a subsequent section! For now, just bear with it through this algorithm:

1. Regardless of orientation, find the cube we need to move into cubicle 7, and perform U until it is in cubicle 3.
2. Repeat the following commutator until the cube in cubicle 3, is oriented correctly in cubicle 7:

$$[R, U] = RUR'U'$$

3. Once complete, perform D' .
4. Repeat steps 2-3 until the bottom layer (white side) is completely solved.
5. Looking at the top layer of cubes, see if they are in the correct cubicles, regardless of their orientations. There will either be 1, 2, or 4 cubes in the correct cubicles. If there are 4, skip to step (8). If there are 1 or



Figure 8: Pocket Cube with facelets labeled.

2, perform the following commutator, performing UD to get a correctly positioned cube into cubicle 4 (i.e., front left):

$$[R, U' L' U] = R U' L' U R' U' L U$$

6. Check again, performing U if necessary to get 1 cubicle in the correct position.
7. Repeat steps 5-6 until all 4 cubes are in the correct cubicles, regardless of orientation.
8. Flip the entire Pocket Cube the solved white (Down) face to the Up face. This is equivalent to performing:

$$R R L L$$

9. Since white is on our Up face, yellow will be our bottom face. Perform D' to get an incorrectly orientated cube into cubicle 7. Then perform the commutator from (2) until it is correctly orientated:

$$[R, U] = R U R' U'$$

10. Perform D' to move the newly solved cube into cubicle 8.
11. Repeat steps 9-10 until the yellow (Down) face is completely solved.
12. Perform U as needed to align the colors on the Front, Back, Left and Right Face.
13. Your Pocket Cube is now solved! [NOB23] [Spe25]

5.3 Why it works!

5.3.1 Commutators.

Commutators are moves in $(G, \cdot)$ that take the form [Swe22]

$$[A, B] = ABA'B'$$

As an example, let's examine the first commutator from the 2x2 Algorithm:

$$[R, U] = RUR'U'$$

There's only 2 faces that are being permuted by $[R, U]$: R and U . We move both, and then move both "back". These faces have one cubicle in common: cubicle 3 (see Figure 9). But, then performing $R'U'$, also permutes cubicles 1 and 2. So, we can see that if we repeatedly perform the commutator $[R, U]$, cubicles 1, 2, and 3 cycle and rotate, while the rest of the cube remains unchanged [Swe22].

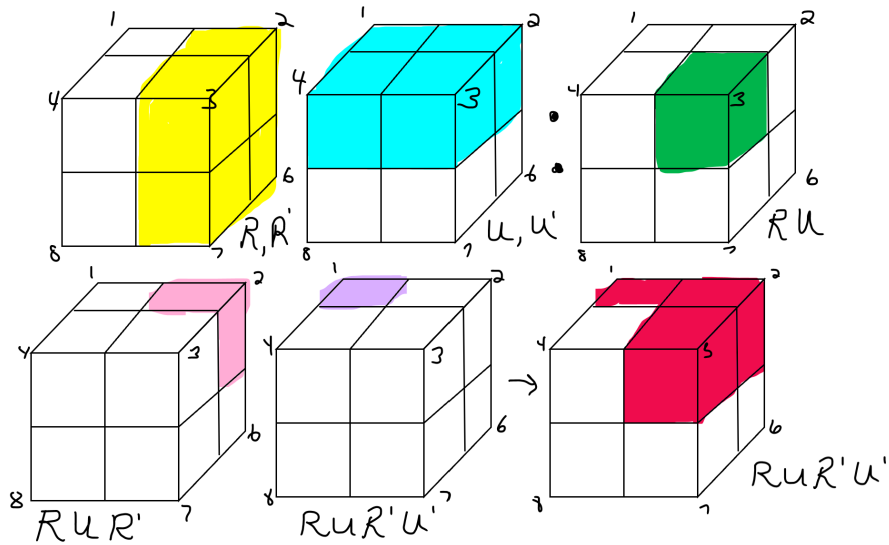


Figure 9: Cubes permuted by both $[R, U]$.

In short, the first half of the commutator $[A, B] = ABA'B'$ – that is AB – will overlap on some subset of the cublets, K . Then, when we "undo" the moves in the second half – $A'B'$ – we change all the cublets back to their original cubicle – except for those in K . So, we can effectively work with only a subset of the cublets, without affecting the rest of the cube.

5.3.2 Conjugates.

Conjugates in the Rubik's Cube world mean exactly the same thing in group theory. They are moves that take the form: z , and x and x' are a set up

move, and the opposite of that set up move. These are useful for using the same commutator to affect a different area of the cube [Swe22].

5.3.3 Symmetric Groups & K-Cycles.

As touched on in the section on Normal Subgroups, the moves $g \in G$ can be represented by bijective maps in the form of permutations.

We can write this in more detail using symmetric groups and cycles. The 6 basic moves can be represented with the following cycles, using the facelet notation [Lau03][Mula]:

- $R = (13, 14, 16, 15)(10, 2, 19, 22)(12, 4, 17, 24)$
- $L = (5, 6, 8, 7)(3, 11, 23, 18)(1, 9, 21, 20)$
- $U = (1, 2, 4, 3)(9, 5, 17, 13)(10, 6, 18, 14)$
- $D = (21, 22, 24, 23)(11, 15, 19, 7)(12, 16, 20, 8)$
- $F = (9, 10, 12, 11)(3, 13, 22, 8)(4, 15, 21, 6)$
- $B = (17, 18, 20, 19)(1, 7, 24, 14)(2, 5, 23, 16)$

or in cubicle notation

- $R = (2, 6, 7, 3)$
- $L = (1, 4, 8, 5)$
- $U = (1, 2, 4, 3)$
- $D = (5, 6, 7, 8)$
- $F = (3, 7, 8, 4)$
- $B = (2, 1, 5, 6)$

Using these cycles, we can use the Python SymPy code [Tea25] in Appendix A to calculate the cycles in the

$$(1)[R, U] = RUR'U' = (1, 14, 18, 17, 5, 2)(4, 15, 13, 22, 10, 12)$$

or

$$(2)[R, U] = RUR'U' = (1, 2)(3, 7) = (1, 2)(3, 4)(4, 5)(5, 6)(6, 7)(5, 6)(4, 5)(3, 4)$$

So, we can see from (2) using this commutator cubicles 1 and 2 continually switch cubes, and so do cubicles 3 and 7 (These are 2-cycles). We can also see from (1) how the facelets swap as the cubicles switch location and rotate, in 6-cycles. It is also important to notice that cubicles 4, 5, 6, and 8 and their respective facelets are completely unaffected by this cycle— which is obviously

useful when trying to solve one layer of the Pocket Cube at a time. From Lauritzen, we also know that $n(\sigma)$ is the number of simple transpositions of which σ is a product, where the simple transposition is of the form $s = (i \ i+1)$. So,

$$n([R, U]) = 8$$

so

$$\text{sgn}([R, U]) = (-1)^{n([R, U])} = +1$$

So $[R, U]$ is an even permutation. And, by Lauritzen we know

$$|[R, U]| = \text{lcm}(6, 6) = 6$$

We can do the same thing for our other commutator $[R, U'L'U]$:

$$(3)[R, U'L'U] = (2, 3)(7, 8)$$

$$(4)[R, U'L'U] = (1, 13, 14)(2, 5, 10)(4, 17, 18)$$

Again, in terms of cubelets, it is the product of two 2-cycles. When looking at the facelets we have the product of three 3-cycles. So, cubicles' 2 and 3, and 7 and 8 are continually swapping cubes and reorienting the cubelet within the cubicle each cycle. Also,

$$|[R, U'L'U]| = \text{lcm}(3, 3, 3) = 3$$

and since $(2, 3)(7, 8)$ are already simple transpositions

$$n([R, U'L'U]) = 2$$

$$\text{sgn}([R, U'L'U]) = (-1)^{n([R, U'L'U])} = +1$$

so it is an even permutation

6 How to Make an Impossible Rubik's Cube

Finally, we have all the information we need to make our impossible Rubik's Cubes. All we need to do is perform an illegal "move" (obviously on a physical Rubik's Cube we cannot do this without removing and rearranging the cubes or the stickers on the facelets, but suspend your disbelief here), and then all solving algorithms will fail. All our illegal move needs to do is violate any one of the three criteria in the 3x3 Rubik's Cubes FTCT. Any Move that violates the v criteria is also an illegal move for the Pocket Cube. So, some illegal moves we could use to make our impossible Rubik's Cube is:

1. Rotate any 1 corner cube.
2. Rotate any 1 edge cube.

3. Swap any 2 adjacent cubes.

To begin, let's examine and verify that $[R, U]$ from above is legal move and justify it. Looking at the Pocket Cube FTCT, the only criteria we need to meet is

$$v_1 + v_2 + \dots + v_8 = 0 \pmod{3}$$

From Figures 10 and 11 we can calculate this quantity.



Figure 10: Cube orientations on Up face after $[R, U]$.

Summing from cubicle 1 to 8:

$$v_1 + v_2 + \dots + v_8 = 0 \pmod{3} = 0 + 1 + 1 + 0 + 0 + 0 + 1 + 0 \pmod{3} = 3 \pmod{3} = 0$$

So, we can tell this is a legal move on the Pocket Cube via the Pocket's Cube FTCT.

Now, examining the illegal moves, Illegal move (1) Rotating any one corner cube is a very simple option. Without loss of generality to all legal positions, consider the solved Rubik's Cube configuration. Here, we have

$$v_1 + v_2 + \dots + v_8 = 0 \pmod{3} = 0 + 0 + 0 + 0 + 0 + 0 + 0 + 0 \pmod{3} = 0$$

Rotating any one corner cube would result in, without loss of generality to the cubicle of the corner we're working with

$$v_1 + v_2 + \dots + v_8 = 0 \pmod{3} = 1 + 0 + 0 + 0 + 0 + 0 + 0 + 0 \pmod{3} = 1 \pmod{3} \neq 0$$

or

$$v_1 + v_2 + \dots + v_8 = 0 \pmod{3} = 2 + 0 + 0 + 0 + 0 + 0 + 0 + 0 \pmod{3} = 2 \pmod{3} \neq 0$$



Figure 11: Cube orientations on Down face after $[R, U]$.

Clearly, this violates the conservation of corner rotations. Rotating more than 1 corner might accidentally result in a sum that is congruent to 0 (mod 3), so best to only rotate! Illegal move (2) works the same way, but instead violates the conservation of edge flips, and is only applicable to the 3x3 Rubik's Cube.

Illegal move (3) is a little more interesting, but is only applicable to the 3x3 Cube. Obviously, we cannot swap a corner with an edge cube, because they have a different number of facelets. Let's find ρ and θ in terms of disjoint cycle notation, in the case where the 2 pieces swapped are corner cubes in adjacent cubicles i and $i + 1$:

$$\rho = (i, i + 1)$$

which consists of 1 simple transposition. Because no edge pieces are involved in this swap we have

$$\theta = (1)(2)(3)(4)(5)(6)(7)(8)(9)(10)(11)(12)$$

which has 0 simple transpositions. Therefore we have

$$\begin{aligned} \text{sgn}(\rho) &\neq \text{sgn}(\theta) \\ (-1)^{n(\rho)} &\neq (-1)^{n(\theta)} \\ (-1)^1 &\neq (-1)^0 \\ -1 &\neq 1 \end{aligned}$$

which violates the FTCT criteria on Equal Parity. Therefore, this is an illegal move, and will render the Rubik's Cube unsolvable.

So, if we are handed a Rubik's Cube, by simply finding the 4-tuple representing it's configuration we can easily determine if it is solvable!

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